

Stage 3 Judges' Rubric - DECI Level 3

ESP32 + Track Integration Project - Stage 3 Judge Evaluation

Team Name / #	Project Name	Judge Name	Date

How to Score Every Row Enter ONE number from 1 to 5. Every row is worth 5 points; harder sections have more rows and therefore more weight.

1	Weak	Missing, incorrect, non-functional, or little/no meaningful evidence.
2	Developing	Partially demonstrated; important gaps remain or substantial support is needed.
3	Meets Expectations	Core requirement is present, understandable, and works at an acceptable/basic level.
4	Strong	Complete, accurate, well integrated, and supported by good evidence; only minor issues.
5	Excellent	Highly effective, reliable, purposeful, well-tested/polished, and clearly justified by the team.

Judge the final Stage 3 work, using the previous-phase requirements as the baseline. **Flow:** reward evidence, correct integration, and purposeful design - not complexity for its own sake. **Testing:** look for real trials/results, not only claims. **Hardware:** reward the best-fit use of components and reliable integration. **Teamwork & presentation:** look for shared ownership, clear communication, and the ability of multiple members to answer questions.

Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Problem, Research & Solution			Section Weight: 15 points
Problem, user & research	Clearly defines the problem/user and supports it with relevant research findings and references.		
Solution & user interaction	Solution directly addresses the problem and explains a complete user interaction.		
Features & value	Main functions are coherent, useful, and consistent with the same target user.		
System, Hardware, IoT & Circuit			Section Weight: 25 points
ESP32 system architecture	System clearly explains sensor/input -> ESP32 processing -> decision -> output/communication, including relevant ESP32 capabilities.		

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Continue using the same 1-5 scale. Enter one score for each specific judging task.

Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
System, Hardware, IoT & Circuit (continued)			Section Weight: 25 points
Purposeful platform & component choice	Kit/platform, ESP32/ESP32 Camera, sensors, and actuators are selected because they fit the project.		
IoT communication	Wi-Fi/Bluetooth/data flow is technically sensible, including two-way communication where needed.		
Circuit correctness	Circuit connections and selected GPIO/input types are reasonable and match the implemented plan.		
Best hardware use & reliability	Hardware/ESP32 capabilities are used effectively and operate consistently under repeated demonstrations.		
Programming & AI / Data			Section Weight: 20 points
Programming/control logic	Inputs, decisions, timing/loops, outputs, and communication are implemented logically and reliably.		
Data collection & decisions	Collected sensor/image data is relevant and is used appropriately to drive decisions or actions.		
AI approach & relevance	If AI/computer vision is used, classification/object detection or other approach fits the problem and data.		
Evidence of AI/data flow	Shows model/data-flow/visual evidence and explains how results connect back to the system.		

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Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Digital Identity & Track Integration			Section Weight: 10 points
Digital identity	Project name, logo, palette, and visual style are consistent and fit the product/user.		
Applied track integration	Relevant team track output (Web, UI/UX, Data Science, Cybersecurity) is clearly connected to the same ESP32 project and shows real evidence.		
Testing & Validation			Section Weight: 10 points
Testing evidence & success checks	Shows repeated functional tests or data/communication tests with observable success criteria.		
Improvement & system reliability	Team identifies weaknesses and improves hardware, logic, data, interface, or communication based on evidence.		
Business & Value			Section Weight: 5 points
Customer, value & pricing	Target customer, value proposition, and basic pricing/delivery idea are clear and realistic.		
Teamwork & Presentation			Section Weight: 15 points
Cross-track teamwork	Members integrate their track contributions into one solution instead of presenting disconnected pieces.		
Communication & technical ownership	Different members can explain decisions and demonstrate understanding beyond only their own slide/part.		

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Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Teamwork & Presentation (continued)			Section Weight: 15 points
Presentation, demo & Q&A;	Project story is clear and professional; evidence is well selected; team handles judge questions effectively.		

Extra Requirements / Optional Bonus			Optional Bonus - max 10
Advanced AI feature	Optional advanced AI/ML feature goes beyond the core data plan and includes prototype/evidence.		
Advanced system feature	Optional stronger IoT, cybersecurity, data processing, or digital experience feature improves the same system and includes evidence.		

Score Summary

Section	Maximum	Section Subtotal	Notes
Problem, Research & Solution	15		
System, Hardware, IoT & Circuit	25		
Programming & AI / Data	20		
Digital Identity & Track Integration	10		
Testing & Validation	10		
Business & Value	5		
Teamwork & Presentation	15		
CORE TOTAL	100		

OPTIONAL BONUS (record separately, max 10)	
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